TRANSTITLE

B1 AND B2

Abstract. abbbaaaaaaaa

1. Introduction

2. Model

3. Introduction and model formulation

3.1. **Physical origin and governing equations.** The transition from laminar to turbulent pipe flow has been investigated by Barkley and collaborators using a reduced model coupling turbulence intensity to local centerline velocity. The original program was initiated in [1] and developed into the model used to describe the rise of fully turbulent flow in [3, 2].

The spatially homogeneous reaction dynamics associated with the later model are

$$\begin{aligned} \dot{q} &= q\big[r + u - 2 - (r + 0.1)(q - 1)^2\big], \\ \dot{u} &= \varepsilon\big[2 - u + 2q(1 - u)\big], \end{aligned} \tag{1}$$

where $q \geq 0$ represents turbulence intensity, u is the centerline-velocity variable, $r > 0$ is a Reynolds-number-related model parameter, and $\varepsilon > 0$ is a timescale parameter. Equations (1) arise by suppressing the spatial-derivative terms in the published reaction–diffusion–advection model; they are not a new artificial modification [3].

References

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